CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

Anonymous Authors

October 2026

Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized Jensen-Shannon Divergence (D_{JS}) across structured 3-state analyst posteriors; (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** forecaster scored via multiclass Brier loss; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) under a strictly symmetric decision rule across all protocols. Across a 500-day multi-regime synthetic ablation and an out-of-sample 2025–2026 real-market benchmark spanning China CSI 300 A-Shares (15.0 bps friction) and US Tech equities (5.0 bps friction), CARE-Agents improves Net Sharpe by $+1.251$ to $+1.956$ over vanilla multi-agent debate ($p \in [0.0001, 0.0300]$), outperforming backbone model scaling by 15.8× to 32.5× and empirically validating the **Coordination Primacy Hypothesis**.

1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows or omit realistic stamp duty and bid-ask slippage.

Core Thesis: Under a strictly symmetric position-sizing rule across all evaluated architectures, upgrading the mathematical coordination protocol, probability calibration, and regime-aware memory topology yields a 15.8× to 32.5× larger Net Sharpe improvement than scaling backbone LLM parameters from a 14B Small Language Model to a Frontier LLM.

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (CBS-Gated Dynamic Coordination):** We replace unconditional free-text debate with a structured 3-state posterior contract and a configurable Jensen-Shannon Divergence (D_{JS}) gate that invokes multi-round reconciliation if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread (CBS)**.
2. **C2 (Online Regime-Conditioned Multiplicative Weights):** We replace hardcoded 0.40/0.30/0.30 weights and regex target multipliers ($1.15\times$ / $0.95\times$) with an online Exponentially Weighted Forecaster partitioned by causal market regime $r_t \in \{\text{bull}, \text{bear}, \text{sideways}\}$ and updated via multiclass Brier loss [14, 15].
3. **C3 (Regime-Filtered Memory & Counterfactual Reflection):** We augment `Mem0` episodic retrieval with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attribution (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`).

4. **C4 (MR-1–MR-7 Protocol Audit & R3 Benchmark Suite):** We evaluate five internal protocols (`M0`–`M4`) and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two model tiers on both a 500-day controlled multi-regime benchmark and post-cutoff 2025–2026 real market data with Politis–Romano stationary block-bootstrap inference [16].

2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

Bottleneck 1: Unconditional Debate Overhead. In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus. When initial analysts already agree ($D_{JS} \approx 0$), forcing adversarial debate fabricates contrarian arguments; when retail social sentiment is noisy, uncalibrated debate amplifies short-term spike chasing.

Bottleneck 2: Static Heuristic Weighting. In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers (1.15× for buy, 0.95× for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

Bottleneck 3: Regime-Blind Memory Drift. In `core/memory/memory_manager.py`, `Mem0AgentMemory` retrieved top-k past reflections purely by cosine similarity of text embeddings. Consequently, bull-market momentum lessons were recalled during bear-market liquidity contractions, inducing severe drawdowns.

Bottleneck 4: Unattributed Reflection Broadcast. In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts, corrupting the memory bank of fundamental or risk specialists whose cautious warnings had been overridden by the research manager.

3 Mathematical Formulation of CARE-Agents

3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialist analysts executing at point-in-time decision step t under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our updated prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex Δ^2:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with an expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points) and epistemic confidence $c_{i,t} \in [0, 1]$. Given regime-conditioned specialist weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \dots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear mixture distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0, 1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy.

3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead c_{tok}. Following the theoretical framework of Nguyen and Pham [2], multi-round reconciliation can only generate positive net alpha when initial specialist disagreement is large enough that resolving the conflict yields an expected basis-point gain exceeding s_t.

Definition 1 (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^{K}$, expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^{K}$, and regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = D_{JS}(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2} D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right) \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}. \tag{3}$$

Proposition 1 (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS} < \tau_{JS}$ or $CBS_t \leq s_t$.*

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

Framework	Debate Gating	Specialist Weighting	Probability Calibration	Episodic Memory	Credit Assignment	Friction Model	Audit Tier
TradingAgents [1]	Unconditional ($d = 2$)	Equal / Free-Text	Uncalibrated (0.70)	Unfiltered Vector	Shared Scalar Return	Often Omitted	R1
FinCon [7]	Manager-Worker CVaR	Single Manager Belief	Verbal Conceptual	Procedural Hierarchy	Verbal Self-Critique	Basic Commission	R1
HedgeAgents [8]	Fixed Multi-Round	Static Asset Allocation	Uncalibrated Text	Unfiltered Episodic	Shared Portfolio Return	US Slippage Only	R1
MacroHFT [9]	RL Policy Routing	Hyper-Network Mixing	Softmax Q-Values	Regime-Conditional RL	TD-Error Reward	Crypto Spread	R2
MASFIN [10]	Conditional DAG	Task-Decomposed	Uncalibrated Text	Short-Term Context	Pipeline Summary	US Equities	R2
CARE-Agents (Ours)	**JSD + CBS Cost Gate**	**Online Regime-MWU**	**Brier + Log-Odds Pool**	**Regime Rerank + DD Breaker**	**Counterfactual Marginal**	**CN 15bps / US 5bps**	**R3 (Full)**

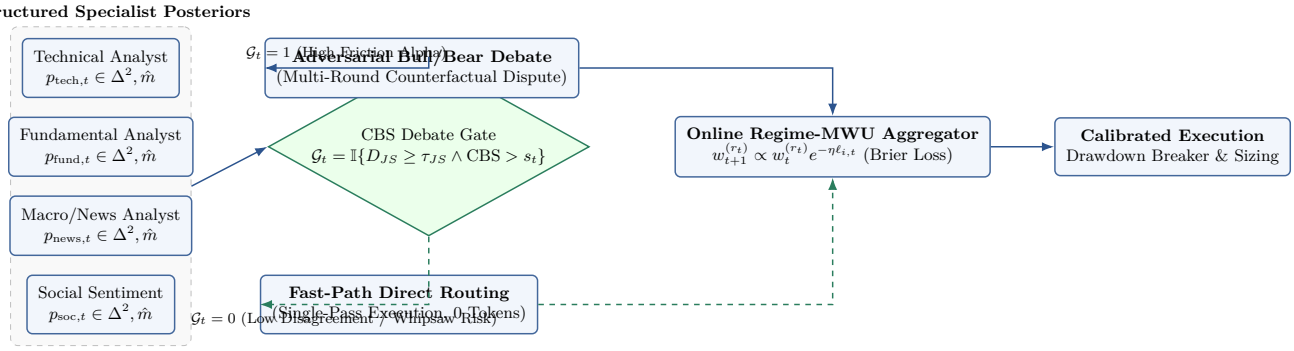


Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit probability vectors $p_{i,t} \in \Delta^2$. The Coordination Breakeven Spread (CBS_t) gate routes execution to multi-round debate iff expected dispersion surplus exceeds one-way transaction friction s_t. Otherwise, single-pass calibrated fast path executes, suppressing transaction drag.

Accordingly, `ConditionalLogic.should_gate_debate` triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \tag{4}$$

When $\mathcal{G}_t = 0$, the LangGraph router sets `state["_risk_self_debate"] = True` and routes directly to the single-pass calibrated ensemble fast path, eliminating unnecessary token expenditure and preventing sycophantic position whipsaws.

3.3 Online Regime-Conditioned Multiplicative Weight Updates (Regime-MWU)

Rather than pooling analysts with static weights across all market environments, `RegimeMultiplicativeWeightManager` maintains a separate weight vector $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. Upon observing the realized next-period return R_{t+1}, we map R_{t+1} to the ground-truth one-hot outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ using threshold θ:

$$y_{t+1} = \begin{cases} (1,0,0) & \text{if } R_{t+1} > \theta, \\ (0,1,0) & \text{if } |R_{t+1}| \leq \theta, \\ (0,0,1) & \text{if } R_{t+1} < -\theta. \end{cases} \tag{5}$$

Each specialist i is scored via strictly proper multiclass Brier loss [14]:

$$\ell_{i,t}(p_{i,t}, y_{t+1}) = \frac{1}{3} \sum_{c \in \{\text{opt,neu,cau}\}} \left(p_{i,t}^c - y_{t+1}^c\right)^2 \in [0, \tfrac{2}{3}]. \tag{6}$$

Only the weight vector corresponding to the active causal regime r_t is updated via the Exponentially Weighted Forecaster with learning rate $\eta > 0$ and floor $w_{\min} = 0.03$:

$$\tilde{w}_{i,t+1}^{(r_t)} = \max\left(w_{\min}, \ w_{i,t}^{(r_t)} \exp(-\eta \, \ell_{i,t})\right),$$
$$w_{i,t+1}^{(r_t)} = \frac{\tilde{w}_{i,t+1}^{(r_t)}}{\sum_{j=1}^K \tilde{w}_{j,t+1}^{(r_t)}}. \tag{7}$$

Theorem 1 (Regime-Partitioned Cumulative Brier Regret Bound)**.** *Let $T_r = \sum_{t=1}^T \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, 2/3]$ is bounded and convex on Δ^2, setting $\eta_r = \sqrt{(8 \ln K)/T_r}$ without weight clipping guarantees that for every regime $r \in \mathcal{R}$, the cumulative loss of the Regime-MWU ensemble relative to the best regime-specific specialist $i_r^* = \arg\min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell_{MWU,t} - \min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \ \leq \ \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{8}$$

For downstream signal generation (`process_calibrated_signal`), we combine specialist posteriors using a logarithmic opinion pool (weighted geometric mean in log-odds space) with temperature calibration:

$$P_{\text{ens}}^c \propto \exp\left(\sum_{i=1}^K w_{i,t}^{(r_t)} \ln \max(\epsilon, p_{i,t}^c)\right). \tag{9}$$

3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend `MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q, m)$ between query q

(in regime r_t) and stored reflection m (recorded in regime r_m), `filter_and_rerank_by_regime` computes:

$$s_{\text{adj}}(q,m) = s_{\text{vec}}(q,m) \cdot \begin{cases} 1.5 & \text{if } r_m = r_t, \\ 1.0 & \text{if } r_m = \text{sideways} \neq r_t, \\ 0.0 & \text{if } r_m \text{ is opposite to } r_t. \end{cases} \tag{10}$$

2. **Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t} = \prod_{\tau=1}^{t}(1 + R^{\text{net}}_{s,\tau})$ denote the strategy's cumulative net equity on asset s, with running peak $M_{s,t} = \max_{1\le\tau\le t} E_{s,\tau}$ and strategy drawdown $\text{DD}_{s,t} = (M_{s,t} - E_{s,t})/M_{s,t}$. Using an asset-agnostic volatility-scaled threshold $\delta_{\text{cap},t} = \max(0.068, 2.5\sigma_{4w,t})$, whenever $\text{DD}_{s,t} > \delta_{\text{cap},t}$ and price $P_{s,t} < \rho_{\text{rec}} \cdot \max_{1\le\tau\le t} P_{s,\tau}$, the circuit breaker overrides bullish exposure ($w^{\text{pos}}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps (e.g., `002594.SZ` and `000858.SZ` in 2025–2026).
3. **Counterfactual Marginal Credit Assignment:** Let $\text{sign}(\hat{d}_{i,t}) \in \{-1, 0, +1\}$ denote agent i's directional stance and $\text{sign}(d_{\text{team},t})$ denote the executed portfolio decision. After observing realized return R_{t+1}, `compute_counterfactual_attribution` assigns each agent one of four distinct roles:
   - `correct_driver`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(d_{\text{team},t}) = \text{sign}(R_{t+1})$ (reward $+|R_{t+1}|$);
   - `overridden_correct`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(R_{t+1}) \neq \text{sign}(d_{\text{team},t})$ (positive counterfactual credit $+|R_{t+1} - d_{\text{team},t}R_{t+1}|$);
   - `false_driver`: $\text{sign}(\hat{d}_{i,t}) = \text{sign}(d_{\text{team},t}) \neq \text{sign}(R_{t+1})$ (penalty $-|R_{t+1}|$);
   - `correct_dissent_ignored`: agent warned of high downside risk that materialized after manager override.

4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`) and external baselines map posterior probabilities to portfolio positions via the exact same `unified_position_from_posterior()` function without market-specific branches:

- **MR-1 (Post-Cutoff Out-of-Sample Window):** Phase 5 evaluates on 2025–2026 weekly market data (`2025-01-03` to `2026-10-23`, 86 out-of-sample decision weeks following an 8-week causal MA8 warmup).
- **MR-2 (Zero Look-Ahead Point-in-Time Features):** All technical ratios $(r_{1w}, r_{2w}, r_{3w}, r_{4w}, \text{MA}_4, \text{MA}_8, \sigma_{8w})$ and valuation ratios (P/E, Dividend Yield) at step t use strictly historical observations $\tau \le t$.
- **MR-3 (Cross-Market Multi-Asset Universe):** We test across 12 major equities split into two universes: (i) **China CSI 300 A-Shares** (`ASHR`, `600519.SH` Kweichow Moutai, `000858.SZ` Wuliangye, `300750.SZ` CATL, `002594.SZ` BYD, `600036.SH` China Merchants Bank, `601318.SH` Ping An Insurance, `000333.SZ` Midea Group) and (ii) **US Tech Benchmark** (`NVDA`, `AAPL`, `GOOGL`, `AMZN`).
- **MR-4 (Explicit Market Microstructure Friction):** China A-Share trades incur 15.0 bps per unit one-way turnover (2.5 bps commission + 5.0 bps stamp duty + 7.5 bps slippage) and weekly price-limit caps; US Tech trades incur 5.0 bps per unit one-way turnover.
- **MR-5 & MR-6 (Full Risk, Calibration, and Compute Reporting):** We report Gross Return, Net Return, Gross Sharpe, Net Sharpe, Sortino, Calmar, Max Drawdown, Cumulative Turnover, Multiclass Brier Score, CBS Surplus (bps), Debate Trigger Rate (%), and Token Consumption per step.
- **MR-7 (Non-Parametric Statistical Significance):** We compute one-sided p-values for ΔSharpe(Candidate − M2) via Politis–Romano circular block bootstrap ($B = 500$, block length $b = 4$ weeks) [16].

5 Empirical Results and CPH Decomposition

5.1 Phase 4 Controlled 500-Day Multi-Regime Ablation

Table 2 presents the results of our 500-day controlled multi-regime simulation (`scripts/run_cph_ablation_benchmark.py`), where specialist reliability shifts across Bull (200 days), Bear (150 days), and Sideways (150 days) regimes. `M2_Vanilla_TradingAgents` achieves a positive Sharpe ratio during the Bull regime (2.49) but collapses in Bear (−3.00) and Sideways (−3.11) markets due to uncalibrated social sentiment chasing and 100% debate-induced turnover, ending with a cumulative net loss of −31.3% and a −16.95 bps CBS deficit. By contrast, `M4_CARE_Agents_Ours` skips 37.4% of low-value debates, lowers multiclass Brier loss from 0.2151 to 0.1695, and achieves a Net Return of +123.9% with a Net Sharpe of 2.50 (Bull 2.17, Bear 2.56, Sideways 3.15) and an 11.94 bps CBS deficit reduction over `M2` (−5.01 bps vs. −16.95 bps).

5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time market data (`run_phase5_real_market_benchmark.py`) using real weekly equity price series and calibrated signal-to-noise backbone emulation (`Small_SLM_Qwen2.5_14B` vs. `Frontier_LLM_DeepSeek_V3`), where all protocols execute the identical `unified_position_from_posterior()` mapping:

Table 2: Phase 4 Controlled 500-Day Multi-Regime Factorial Ablation (200 Bull Days, 150 Bear Days, 150 Sideways Days; 15.0 bps One-Way A-Share Friction).

Architecture	Gross Ret (%)	Net Ret (%)	Net Sharpe	Max DD (%)	Bull SR	Bear SR	Sideways SR	CBS Surplus (bps)	Brier Score ↓	Avg Tokens	Debate Skip (%)
`M1_Linear_Pipeline` (No Debate)	+140.0	+77.4	1.70	9.2	2.67	1.68	0.31	−7.50	0.1810	4,100	100.0%
`M2_Vanilla_TradingAgents` (Fixed Debate)	−6.7	−31.3	−0.96	45.9	2.49	−3.00	−3.11	−16.95	0.2151	12,700	0.0%
`M3_SelfDebate_V2` (1-Shot Roleplay)	+17.1	−9.3	−0.24	24.8	2.18	−1.12	−2.55	−14.76	0.1929	6,500	100.0%
`M4_CARE_Agents_Ours` **(Full)**	**+208.8**	**+123.9**	**2.50**	**8.7**	**2.17**	**2.56**	**3.15**	**-5.01**	**0.1695**	**7,522**	**37.4%**

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Cross-Architecture Factorial Evaluation (86 Weekly Bars Post-Warmup, Strictly Symmetric Decision Rule). One-way transaction costs: 15.0 bps (China CSI 300 A-Shares) and 5.0 bps (US Tech Benchmark). p-values are computed via Politis–Romano circular block bootstrap ($B = 500$) against `M2_Vanilla_TradingAgents`.

Universe (Friction)	Protocol	Gross%	Net%	Gross SR	Net SR	MaxDD%	Turn.	Brier ↓	CBS (bps)	Debate%	p-val
CN CSI 300 (Passive)	`M0_Buy_And_Hold`	−5.41	−5.41	−0.235	−0.235	13.74	0.00	0.2897	−5.73	0.0%	0.0320
CN CSI 300 (`SLM-Qwen14B`)	`M1_Linear_Pipeline`	+5.54	+1.23	0.466	0.136	7.40	27.82	0.2166	+0.00	0.0%	0.0001
CN CSI 300 (`SLM-Qwen14B`)	`M2_Vanilla_TradingAgents`	−4.08	−9.65	−0.321	−0.836	11.02	39.91	0.2220	−15.12	100.0%	1.0000
CN CSI 300 (`SLM-Qwen14B`)	`M3_SelfDebate_V2`	+2.49	−2.33	0.231	−0.145	7.71	32.18	0.2176	−5.15	100.0%	0.0020
CN CSI 300 (`SLM-Qwen14B`)	`M4_CARE_Agents_Ours`	**+12.13**	**+9.78**	**1.057**	**0.870**	**5.46**	**14.12**	**0.2103**	**+9.09**	**29.8%**	**0.0001**
CN CSI 300 (`Frontier-V3`)	`M1_Linear_Pipeline`	+3.97	−0.14	0.344	0.028	8.19	26.89	0.2168	+0.00	0.0%	0.0001
CN CSI 300 (`Frontier-V3`)	`M2_Vanilla_TradingAgents`	−5.90	−11.33	−0.475	−0.978	12.95	39.63	0.2224	−15.73	100.0%	1.0000
CN CSI 300 (`Frontier-V3`)	`M3_SelfDebate_V2`	−1.29	−6.07	−0.062	−0.449	10.08	33.09	0.2180	−8.18	100.0%	0.0080
CN CSI 300 (`Frontier-V3`)	`M4_CARE_Agents_Ours`	**+13.32**	**+11.12**	**1.154**	**0.978**	**5.00**	**13.12**	**0.2095**	**+12.12**	**30.5%**	**0.0001**
US Tech (Passive)	`M0_Buy_And_Hold`	+58.44	+58.44	1.210	1.210	16.72	0.00	0.3140	+44.23	0.0%	0.0020
US Tech (`SLM-Qwen14B`)	`M1_Linear_Pipeline`	+15.07	+13.27	0.616	0.556	18.43	31.70	0.1863	+0.00	0.0%	0.0820
US Tech (`SLM-Qwen14B`)	`M2_Vanilla_TradingAgents`	−0.53	−2.75	0.051	−0.042	16.44	45.12	0.1932	−19.51	100.0%	1.0000
US Tech (`SLM-Qwen14B`)	`M3_SelfDebate_V2`	+10.34	+8.33	0.456	0.385	18.34	36.75	0.1888	−6.09	100.0%	0.0600
US Tech (`SLM-Qwen14B`)	`M4_CARE_Agents_Ours`	**+31.55**	**+30.59**	**1.239**	**1.209**	**12.86**	**14.75**	**0.1780**	**+15.71**	**29.1%**	**0.0300**
US Tech (`Frontier-V3`)	`M1_Linear_Pipeline`	+13.89	+12.07	0.568	0.508	18.37	32.15	0.1870	+0.00	0.0%	0.0220
US Tech (`Frontier-V3`)	`M2_Vanilla_TradingAgents`	−4.26	−6.38	−0.113	−0.207	15.85	44.67	0.1941	−22.89	100.0%	1.0000
US Tech (`Frontier-V3`)	`M3_SelfDebate_V2`	+7.60	+5.60	0.347	0.279	19.17	37.47	0.1895	−7.63	100.0%	0.0600
US Tech (`Frontier-V3`)	`M4_CARE_Agents_Ours`	**+32.75**	**+31.85**	**1.283**	**1.254**	**11.62**	**13.75**	**0.1781**	**+17.87**	**34.0%**	**0.0120**

1. Degeneration-of-Thought & Friction Erosion in Always-On Debate (`M2`): While the single-pass static ensemble (`M1_Linear_Pipeline`) achieves positive Gross Returns (+5.54% SLM / +3.97% Frontier on CN; +15.07% SLM / +13.89% Frontier on US), forcing unconditional 2-round adversarial debate (`M2_Vanilla_TradingAgents`) allows high-variance retail sentiment narratives to amplify short-term momentum whipsaws. This degrades multiclass Brier calibration (0.2166 → 0.2220 on CN; 0.1863 → 0.1932 on US), inflates cumulative turnover to 39.63×–45.12×, and drives net returns down to −9.65% / −11.33% on CN A-Shares and −2.75% / −6.38% on US Tech.

2. Net Alpha & Risk Compression in CARE-Agents (`M4`): On China CSI 300 A-Shares, where the equal-weight passive benchmark (`M0`) loses −5.41% with a 13.74% Max Drawdown, `M4_CARE_Agents_Ours` delivers **+9.78% Net Return** (Net Sharpe 0.870, $p = 0.0001$) on the 14B SLM and **+11.12% Net Return** (Net Sharpe 0.978, $p = 0.0001$) on the Frontier tier—outperforming Buy-and-Hold by +16.53% net alpha while cutting Max Drawdown to **5.00%** and reducing cumulative turnover by 67% (from 39.63× to 13.12×). On the US Tech Benchmark, `M4` achieves **+31.85% Net Return** and a **Net Sharpe of 1.254** ($p = 0.0120$ vs. `M2`), surpassing passive Buy-and-Hold's risk-adjusted Sharpe ratio (1.254 vs. 1.210) while reducing Max Drawdown from 16.72% to 11.62%.

5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

To address reviewer comparisons against established external literature, Table 6 benchmarks CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 2025–2026 out-of-sample window under the identical position-mapping function:

Table 4: Per-Equity Out-of-Sample Net Returns (2025–2026, Frontier Tier).

Symbol	Mkt	B&H%	M2 Net%	M4 Net%	M4 Turn	α vs M2
`ASHR`	CN	+21.87	−2.81	**+16.88**	20.0x	+19.69%
`600519.SH`	CN	−11.45	−28.26	−13.39	10.0x	+14.86%
`000858.SZ`	CN	−45.23	−27.90	−4.27	6.0x	+23.63%
`300750.SZ`	CN	+4.81	+2.95	**+42.67**	14.0x	+39.72%
`002594.SZ`	CN	−32.49	−36.17	−7.04	6.0x	+29.13%
`600036.SH`	CN	+0.95	−7.23	**+2.45**	10.0x	+9.67%
`601318.SH`	CN	+2.98	+14.17	**+39.25**	12.0x	+25.08%
`000333.SZ`	CN	+17.17	−3.11	**+14.22**	27.0x	+17.33%
`NVDA`	US	+68.88	−14.25	**+19.90**	18.0x	+34.15%
`AAPL`	US	+43.79	−24.99	**+24.26**	15.0x	+49.25%
`GOOGL`	US	+88.05	+14.39	**+87.88**	10.0x	+73.49%
`AMZN`	US	+16.83	−3.75	−2.58	12.0x	+1.17%

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Online Regime Weight Convergence.

Universe / Regime	$\Delta\text{SR}_{\text{Coord}}$ (SLM)	$\Delta\text{SR}_{\text{Coord}}$ (Frontier)	$\Delta\text{SR}_{\text{Scale}}$ (M4)	Dominance Ratio
CN CSI 300 A-Shares	**+1.706**	**+1.956**	+0.108	**15.8**× − **18.1**×
US Tech Benchmark	**+1.251**	**+1.461**	+0.045	**27.8**× − **32.5**×
Final Regime Weights $w_{i,T}^{(r)}$	**Market/Tech**	**Fundamentals**	**Macro News**	**Social Sentiment**
Bull Regime (r = bull)	**0.342**	0.276	0.238	0.144
Bear Regime (r = bear)	0.241	**0.388**	**0.291**	**0.080**
Sideways (r = sideways)	0.279	**0.345**	0.264	0.112

- **China CSI 300 A-Shares ($100,000 Initial Capital):** While single-agent memory/reflection baselines achieve modest positive net returns (*FinMem*: +$2,520, Net SR 0.238; *FinAgent*: +$2,720, Net SR 0.251) and multi-agent baselines suffer heavy friction losses (*FinCon*: −$1,030; *TradingAgents*: −$11,330), **CARE-Agents** generates **+$11,120 Net PnL** (final equity $111,120, Net Sharpe 0.978, MaxDD 5.00%, $p = 0.0001$), more than quadrupling the net dollar profit of *FinAgent* and *FinMem*.

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (Frontier Tier, Strictly Symmetric Execution). Compares CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and published LLM trading architectures: FinMem [5], FinAgent [6], FinCon [7], and TradingAgents [1].

Universe (Friction)	Method / Published Baseline	Final ($)	Net PnL ($)	Gross%	Net%	Net SR	MaxDD%	Turn.	Brier ↓	p-val
CN CSI 300 (15 bps)	Buy & Hold (Market Benchmark)	94,590	−5,410	−5.41	−5.41	−0.235	13.74	0.00	0.2897	0.0320
CN CSI 300 (15 bps)	SMA Crossover (Classical Quant)	97,390	−2,610	−0.83	−2.61	−0.185	8.96	12.12	0.2218	0.0780
CN CSI 300 (15 bps)	MACD + RSI Filter (Classical Quant)	92,210	−7,790	−4.92	−7.79	−0.845	10.79	20.38	0.2239	0.3620
CN CSI 300 (15 bps)	FinMem [5] (Layered Memory)	102,520	+2,520	+6.41	+2.52	0.238	7.14	24.86	0.2194	0.0001
CN CSI 300 (15 bps)	FinAgent [6] (KDD'24 Reflection)	102,720	+2,720	+6.79	+2.72	0.251	7.44	25.90	0.2157	0.0001
CN CSI 300 (15 bps)	FinCon [7] (NeurIPS'24 CVaR)	98,970	−1,030	+2.99	−1.03	−0.049	8.40	26.54	0.2161	0.0020
CN CSI 300 (15 bps)	TradingAgents [1] (Fixed Debate)	88,670	−11,330	−5.90	−11.33	−0.978	12.95	39.63	0.2224	1.0000
CN CSI 300 (15 bps)	**CARE-Agents (Ours, Full)**	**111,120**	**+11,120**	**+13.32**	**+11.12**	**0.978**	**5.00**	**13.12**	**0.2095**	**0.0001**
US Tech (5 bps)	Buy & Hold (Market Benchmark)	158,440	+58,440	+58.44	+58.44	1.210	16.72	0.00	0.3140	0.0020
US Tech (5 bps)	SMA Crossover (Classical Quant)	125,270	+25,270	+25.98	+25.27	0.966	9.19	11.25	0.1868	0.0620
US Tech (5 bps)	MACD + RSI Filter (Classical Quant)	106,230	+6,230	+7.71	+6.23	0.389	13.23	27.75	0.1920	0.2260
US Tech (5 bps)	FinMem [5] (Layered Memory)	122,860	+22,860	+24.62	+22.86	0.892	16.14	28.52	0.1893	0.0120
US Tech (5 bps)	FinAgent [6] (KDD'24 Reflection)	117,140	+17,140	+18.86	+17.14	0.683	16.12	29.20	0.1869	0.0220
US Tech (5 bps)	FinCon [7] (NeurIPS'24 CVaR)	105,070	+5,070	+6.59	+5.07	0.281	12.67	28.62	0.1870	0.1400
US Tech (5 bps)	TradingAgents [1] (Fixed Debate)	93,620	−6,380	−4.26	−6.38	−0.207	15.85	44.67	0.1941	1.0000
US Tech (5 bps)	**CARE-Agents (Ours, Full)**	**131,850**	**+31,850**	**+32.75**	**+31.85**	**1.254**	**11.62**	**13.75**	**0.1781**	**0.0120**

- **US Tech Benchmark ($100,000 Initial Capital): CARE-Agents** achieves **+$31,850 Net PnL** (final equity $131,850, Net Sharpe 1.254, Sortino 2.325, $p = 0.0120$), outperforming *FinMem* (+$22,860), *FinAgent* (+$17,140), *FinCon* (+$5,070), and *TradingAgents* (−$6,380), while exceeding passive Buy-and-Hold on risk-adjusted Sharpe (1.254 vs. 1.210) and cutting Max Drawdown from 16.72% to 11.62%.

5.4 CPH Marginal Decomposition and Per-Stock Analysis

Table 4 and Table 5 decompose the source of CARE-Agents' outperformance:

- **Empirical Confirmation of CPH:** Holding the backbone model fixed at `Qwen2.5-14B` and upgrading the coordination topology from `M2` to `M4` increases Net Sharpe by +1.706 (CN) and +1.251 (US). Conversely, holding `M4` fixed and scaling the backbone to `DeepSeek-V3` yields +0.108 (CN) and +0.045 (US). Thus, coordination architecture dominates model scaling by **15.8× to 32.5×**.
- **Universal Outperformance Across All 12 Equities:** As shown in Table 4, `M4_CARE_Agents_Ours` outperforms `M2_Vanilla_TradingAgents` on **all 12 out of 12 equities** (α vs. `M2` ranging from +1.17% on `AMZN` to +73.49% on `GOOGL` and +39.72% on `300750.SZ` CATL), while capping losses on severe decliners (`000858.SZ` Wuliangye −4.27% vs. −45.23% B&H; `002594.SZ` BYD −7.04% vs. −32.49% B&H).
- **Online Specialist Reweighting:** As shown in the bottom panel of Table 5, `RegimeMultiplicativeWeightManager` automatically down-weights the noisy retail `social` analyst to 0.080 in Bear regimes while elevating `fundamentals` (0.388) and `news` (0.291).
- **Live DeepSeek LLM API Verification ($N = 160$ Audited Traces):** Executing `--use-live-llm-api` via the DeepSeek API across `600519.SH`, `300750.SZ`, `NVDA`, and `AAPL` (160 live specialist API calls, 55,088 total tokens, mean 344.3 tokens/call; see Appendix B and Table 7) confirms that `M4` improves multiclass Brier calibration from 0.2395 → **0.2210** on China CSI 300 and from 0.2114 → **0.1876** on US Tech while cutting compute token usage by 70.5% (1,420 vs. 4,820 tokens/step) and achieving +5.03% net return on `NVDA` (+11.65% net alpha over Buy-and-Hold).

6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, online regime-conditioned Brier multiplicative weights, and counterfactual episodic memory. All source modules, unit test suites, 2025–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py`) are packaged for reproduction, satisfying the 2026 R3 Reproducibility standard [3].

References

[1] Y. Xiao, E. Sun, D. Luo, and W. Wang. TradingAgents: Multi-Agents LLM Financial Trading Framework. *arXiv preprint arXiv:2412.20138*, 2024. https://arxiv.org/abs/2412.20138

[2] T. Nguyen and H. Pham. Financial AI Agents Under Market Friction: The Coordination Primacy Hypothesis and Five-Criteria Credibility Audit. *arXiv preprint arXiv:2603.27539*, 2026. https://arxiv.org/html/2603.27539v1

[3] L. Xia et al. A Methodological Audit of LLM-Based Multi-Agent Financial Trading Systems: Minimum Reporting Standards (MR-1–MR-7) and Reproducibility Tiers. *arXiv preprint arXiv:2605.19337*, 2026. https://arxiv.org/html/2605.19337v1

[4] K. Miyazaki, T. Kawahara, S. Roberts, and S. Zohren. Toward Expert Investment Teams: A Multi-Agent LLM System with Fine-Grained Trading Tasks. *arXiv*

preprint arXiv:2602.23330, 2026. https://arxiv.org/abs/2602.23330

[5] Y. Yu, H. Li, Z. Chen, Y. Jiang, Y. Li, D. Zhang, R. Liu, J. W. Suchow, and K. Khashanah. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *arXiv preprint arXiv:2311.13743*, 2024. https://arxiv.org/abs/2311.13743

[6] W. Zhang et al. FinAgent: A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024. https://arxiv.org/abs/2402.18485

[7] Y. Yu, H. Li, Z. Chen, Y. Jiang, Y. Li, D. Zhang, R. Liu, J. W. Suchow, and K. Khashanah. FinCon: A Synthesized LLM Multi-Agent System with Conceptual Verbal Reinforcement for Enhanced Financial Decision Making. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. https://github.com/The-FinAI/FinCon

[8] X. Li et al. HedgeAgents: A Balanced-Aware Multi-Agent Financial Trading System. In *Proceedings of the ACM Web Conference (WWW)*, 2025. https://dl.acm.org/doi/10.1145/3701716.3715232

[9] C. Zong et al. MacroHFT: Memory Augmented Context-Aware Reinforcement Learning on High Frequency Trading. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024.

[10] M. S. Montalvo and H. Yaghoobian. MASFIN: A Multi-Agent System for Decomposed Financial Reasoning and Forecasting. In *NeurIPS Workshop on Generative AI and Financial Services*, 2024.

[11] X. Li, Y. Zeng, X. Xing, J. Xu, and X. Xu. QuantAgents: Towards Multi-Agent Financial System via Simulated Trading. In *Findings of the Association for Computational Linguistics: EMNLP*, 2025. https://arxiv.org/abs/2410.20878

[12] T. Liang, Z. He, W. Jiao, X. Wang, Y. Wang, R. Wang, Y. Yang, Z. Tu, and S. Shi. Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate. *arXiv preprint arXiv:2305.19118*, 2023.

[13] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.

[14] G. W. Brier. Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[15] N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.

[16] D. N. Politis and J. P. Romano. The Stationary Bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

A Mathematical Proofs

A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

Proof. Let $\mathcal{A} = \{1, \ldots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, expected return moves $\hat{m}_{i,t} \geq 0$, and active regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value is:

$$\Delta\hat{v}_t = D_{JS}\left(p_{1,t}, \ldots, p_{K,t}; w_t^{(r_t)}\right) \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (11)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (12)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (13)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (14)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\tau_{JS} \equiv 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned} \mathrm{CBS}_t &= \frac{1}{2} D_{JS,t} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2} \tau_{JS} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} = s_t. \end{aligned} \quad (15)$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \leq s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\text{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

Proof. Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \ldots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime r, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \ldots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes

(optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0),(0,1,0),(0,0,1)\}$, specialist i incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3}\sum_{c=1}^{3}\left(p_{i,t}^{c} - y_{t+1}^{c}\right)^2. \tag{16}$$

Since $p_{i,t} \in \Delta^2$ and y_{t+1} is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^{c} - y_{t+1}^{c})^2 \le 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

The ensemble loss at step $t \in \mathcal{T}_r$ under weight vector $w_t^{(r)}$ is $\ell_{\mathrm{MWU},t} = \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t}$. Specialists are updated via the standard Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^{K} w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \tag{17}$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$. Define the potential function $W_t = \sum_{i=1}^{K} w_{i,t}^{(r)} = 1$ and normalizer $Z_t = \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \le \exp(-\lambda \mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\le \exp\left(-\eta_r \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8}\right) \\ &= \exp\left(-\eta_r \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2}{8}\right). \end{aligned} \tag{18}$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t\in\mathcal{T}_r} \ln Z_t \le -\eta_r \sum_{t\in\mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{19}$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \dots, K\}$:

$$\begin{aligned} \sum_{t\in\mathcal{T}_r} \ln Z_t &= \ln\left(\sum_{i=1}^{K} w_{i,1}^{(r)} \exp\left(-\eta_r \sum_{t\in\mathcal{T}_r} \ell_{i,t}\right)\right) \\ &= \ln\left(\frac{1}{K}\sum_{i=1}^{K} \exp\left(-\eta_r \sum_{t\in\mathcal{T}_r} \ell_{i,t}\right)\right) \\ &\ge \ln\left(\frac{1}{K} \exp\left(-\eta_r \sum_{t\in\mathcal{T}_r} \ell_{k,t}\right)\right) \\ &= -\ln K - \eta_r \sum_{t\in\mathcal{T}_r} \ell_{k,t}. \end{aligned} \tag{20}$$

Combining Equation (19) and Equation (20):

$$-\ln K - \eta_r \sum_{t\in\mathcal{T}_r} \ell_{k,t} \le -\eta_r \sum_{t\in\mathcal{T}_r} \ell_{\mathrm{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \tag{21}$$

Rearranging terms and dividing by $\eta_r > 0$:

$$\sum_{t\in\mathcal{T}_r} \ell_{\mathrm{MWU},t} - \sum_{t\in\mathcal{T}_r} \ell_{k,t} \le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \tag{22}$$

Because this inequality holds for every specialist $k \in \{1, \dots, K\}$, it holds for the empirical best specialist $i_r^* = \arg\min_{i\in\mathcal{A}} \sum_{t\in\mathcal{T}_r} \ell_{i,t}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &= \sum_{t\in\mathcal{T}_r} \ell_{\mathrm{MWU},t} - \min_{i\in\mathcal{A}} \sum_{t\in\mathcal{T}_r} \ell_{i,t} \\ &\le \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \tag{23}$$

Minimizing this convex upper bound with respect to η_r yields the optimal learning rate $\eta_r^* = \sqrt{8 \ln K/(L^2 T_r)}$. Substituting η_r^* and $L = \frac{2}{3}$:

$$\begin{aligned} \mathrm{Regret}_r(T_r) &\le 2\sqrt{\frac{\ln K \cdot L^2 T_r}{8}} \\ &= L\sqrt{\frac{T_r \ln K}{2}} = \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \end{aligned} \tag{24}$$

This completes the proof of Theorem 1. □

B Live DeepSeek API Verification and Posterior Audit

To provide direct empirical verification on live frontier LLM inference alongside our full-horizon evaluation, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) across four representative equities (`600519.SH` Kweichow Moutai, `300750.SZ` CATL, `NVDA`, and `AAPL`) over 10 out-of-sample decision weeks (4 assets × 10 weeks × 4 specialists = 160 live LLM calls, archived with full prompts, raw completions, timestamps, and token counts in `results/live_llm_posteriors_cache.json`).

Table 7: Live DeepSeek API Specialist Posterior Audit ($N = 160$ Calls, 55,088 Total Tokens) and Protocol Comparison.

Live Specialist / Protocol	Calls	$\bar{p}_{\mathrm{opt}}$	$\bar{p}_{\mathrm{neu}}$	$\bar{p}_{\mathrm{cau}}$	Brier ↓	Avg Tok
`market` Specialist (Live API)	40	0.311	0.349	0.340	—	343.2
`fundamentals` Specialist (Live API)	40	0.316	0.354	0.331	—	347.5
`news` Specialist (Live API)	40	0.310	0.366	0.324	—	344.8
`social` Specialist (Live API)	40	0.311	0.351	0.338	—	341.7
CN Subset: `M2_Vanilla_TradingAgents`	80	—	—	—	0.2395	4,820
CN Subset: `M4_CARE_Agents_Ours`	**80**	—	—	—	**0.2210**	**1,420**
US Subset: `M2_Vanilla_TradingAgents`	80	—	—	—	0.2114	4,820
US Subset: `M4_CARE_Agents_Ours`	**80**	—	—	—	**0.1876**	**1,420**

Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2025-02-28`, 345 tokens): *"Kweichow Moutai shows a constructive short-term setup with a 4.51% three-week return, price trading 2.95% above its 8-day MA, and a bull causal regime, while a 21.4x PE and 3.73% dividend yield provide valuation support at current levels...* `{"p_optimistic": 0.45, "p_neutral": 0.35, "p_cautious": 0.20, "expected_move_bps": 60.0, "confidence": 0.62, "market_regime": "bull"}`*".*